Linear Algebra and Its Applications

Section 4.4: Coordinate Systems

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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4.4 Coordinate systems

An ordered basis $\mathcal{B} = (b_1, \ldots, b_n)$ assigns to each vector x its unique list of weights $[x]_{\mathcal{B}} = (c_1, \ldots, c_n)^T$ in the expression $x = \sum_j c_j b_j$. Changing the basis order changes the meaning of this list.

For a basis in $\mathbb{R}^n$, put $P_{\mathcal{B}} = [b_1 \ \cdots \ b_n]$. The two directions are

$$x = P_{\mathcal{B}}[x]_{\mathcal{B}}, \qquad [x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x.$$

The first assembles a vector from weights; the second finds its weights. For polynomials use the standard coefficient order $1, t, t^2, \ldots$. A missing power has coefficient zero and must still occupy its position.

Practice problem 4.4.P1: Both directions of a basis-coordinate matrix

(a) Check independence and spanning.

The matrix

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

is triangular with three nonzero pivots. It is invertible, so its columns are independent and span $\mathbb{R}^3$; they are a basis.

(b)–(c) State the matrix and direction correctly.

The displayed matrix is the change from $\mathcal{B}$ coordinates to standard coordinates, and the equation is $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$.

(d) Solve for the weights.

For $x = (-8, 2, 3)^T$, the last equation is $3c_3 = 3$, so $c_3 = 1$. The middle equation $4c_2 - 6c_3 = 2$ gives $c_2 = 2$. The first $c_1 - 3c_2 + 3c_3 = -8$ gives $c_1 = -5$.

Conclusion. $[x]_{\mathcal{B}} = (-5, 2, 1)^T$, and multiplication by the matrix gives back $(-8, 2, 3)^T$.

Practice problem 4.4.P2: Coordinates of a quadratic polynomial

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & -1 & 1 & -3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 5 \\ c_2 = 1 \\ c_3 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 1 \\ -2 \end{bmatrix}$.

Exercise 4.4.1: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 3 & -4 \\ -5 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (3)(5) + (-4)(3) \\ (-5)(5) + (6)(3) \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.2: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 3 & -4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -26 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (3)(-2) + (-4)(5) \\ (2)(-2) + (1)(5) \end{bmatrix} = \begin{bmatrix} -26 \\ 1 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.3: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 1 & 5 & 4 \\ -2 & 0 & -3 \\ 3 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (1)(1) + (5)(0) + (4)(-2) \\ (-2)(1) + (0)(0) + (-3)(-2) \\ (3)(1) + (-2)(0) + (0)(-2) \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.4: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} -2 & 3 & 4 \\ 2 & 0 & -1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (-2)(-3) + (3)(2) + (4)(-1) \\ (2)(-3) + (0)(2) + (-1)(-1) \\ (0)(-3) + (2)(2) + (3)(-1) \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.5: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 3 & -1 \\ -2 & -5 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 3 & -1 \\ 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 2 \\ c_2 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Exercise 4.4.6: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ -4 & -3 & -6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-4)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 5 & -10 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{5})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

Exercise 4.4.7: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ -3 & 9 & 4 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ -3 & 9 & 4 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 10 & 30 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{10})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = -1 \\ c_2 = -1 \\ c_3 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}$.

Exercise 4.4.8: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 3 & 8 & 3 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 3 & 8 & 3 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 2 & 0 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 1 \\ c_2 = -1 \\ c_3 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Exercise 4.4.9: Coordinates and basis representations

The map from basis coordinates to standard coordinates places each ordered basis vector in the corresponding matrix column:

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -3 & -5 \end{bmatrix}.$$

Indeed, multiplying by $(c_1, \ldots, c_n)^T$ gives $c_1 b_1 + \cdots + c_n b_n = x$, so this matrix has exactly the required direction.

Exercise 4.4.10: Coordinates and basis representations

The map from basis coordinates to standard coordinates places each ordered basis vector in the corresponding matrix column:

$$P_{\mathcal{B}} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & -2 \\ 6 & -4 & 3 \end{bmatrix}.$$

Indeed, multiplying by $(c_1, \ldots, c_n)^T$ gives $c_1 b_1 + \cdots + c_n b_n = x$, so this matrix has exactly the required direction.

Exercise 4.4.11: Coordinates and basis representations

Invert the ordered basis matrix.

For a 2×2 matrix, the inverse is the adjugate divided by the nonzero determinant. Here

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -3 \\ -2 & 5 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = -1, \qquad P_{\mathcal{B}}^{-1} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix}.$$

Apply the inverse to the actual standard vector.

$$[x]_{\mathcal{B}} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Multiplying the answer by $P_{\mathcal{B}}$ returns x, so the direction of the change has been checked.

Exercise 4.4.12: Coordinates and basis representations

Invert the ordered basis matrix.

For a 2×2 matrix, the inverse is the adjugate divided by the nonzero determinant. Here

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = 1, \qquad P_{\mathcal{B}}^{-1} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}.$$

Apply the inverse to the actual standard vector.

$$[x]_{\mathcal{B}} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 5 \end{bmatrix}.$$

Multiplying the answer by $P_{\mathcal{B}}$ returns x, so the direction of the change has been checked.

Exercise 4.4.13: Coordinates and basis representations

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 1 & 1 & 1 & 7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 0 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 2 \\ c_2 = 6 \\ c_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$.

Exercise 4.4.14: Coordinates and basis representations

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ -1 & -1 & 1 & -6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & -1 & 3 & -5 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 2 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = 2 \\ c_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$.

Exercise 4.4.15: Coordinates and basis representations

(a) True. A basis with n ordered members produces exactly n real weights, hence a coordinate vector in $\mathbb{R}^n$, whatever type of object x itself may be.

(b) False. For a basis of $\mathbb{R}^n$, $P_{\mathcal{B}}$ maps weights to standard vectors: $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$. The coordinate map uses $P_{\mathcal{B}}^{-1}$ instead.

(c) False. The four independent polynomials $1, t, t^2, t^3$ would have independent images under an isomorphism, but four vectors cannot be independent in $\mathbb{R}^3$. In fact $\mathcal{P}_3$ corresponds to $\mathbb{R}^4$.

Exercise 4.4.16: Coordinates and basis representations

(a) True. For the standard basis, $x = x_1e_1 + \cdots + x_ne_n$, so its basis weights are exactly its usual coordinates.

(b) False. The coordinate mapping goes from x to $[x]_{\mathcal{B}}$. The displayed reverse direction is its inverse, assembling a vector from its weights.

(c) True. Every two-dimensional plane through the origin has a two-vector basis. Its coordinate map is a linear bijection onto $\mathbb{R}^2$. An affine plane not through zero would not be a vector space under the usual operations.

Exercise 4.4.17: Coordinates and basis representations

One representation is $5v_1 - 2v_2 + 0v_3 = (1, 1)^T$. There is also the nontrivial relation $5v_1 - v_2 + v_3 = 0$, since

$$5(1, -3)^T - (2, -8)^T + (-3, 7)^T = (0, 0)^T.$$

Add that zero relation to the first representation. It gives $10v_1 - 3v_2 + v_3 = (1, 1)^T$. The two weight triples $(5, -2, 0)$ and $(10, -3, 1)$ differ, and direct substitution verifies both.

Conclusion. The failure of independence permits distinct representations of the same vector.

Exercise 4.4.18: Coordinates and basis representations

The jth basis vector has the representation

$$b_j = 0b_1 + \cdots + 0b_{j-1} + 1b_j + 0b_{j+1} + \cdots + 0b_n.$$

A basis makes representations unique, so these are exactly its coordinates. The coefficient column is e_j, with a 1 in place j and zero elsewhere.

Conclusion. $[b_j]_{\mathcal{B}} = e_j$ for each j, the columns of I_n.

Exercise 4.4.19: Coordinates and basis representations

The assumed existence of a representation for every vector says Span $S = V$. To establish independence, compare two representations of zero. One uses all coefficients zero. If a nontrivial zero combination existed, it would give a different representation of zero, contradicting uniqueness. Thus the only zero combination is trivial.

Conclusion. S is independent and spans V, hence is a basis.

Exercise 4.4.20: Coordinates and basis representations

Because the four generators span, write $w = \sum_{j=1}^{4} k_j v_j$. Their dependence supplies $a_1, \ldots, a_4$, not all zero, with $\sum_j a_j v_j = 0$. For every real t,

$$w = \sum_{j=1}^{4} (k_j + t a_j) v_j.$$

At least one a_j is nonzero, so changing t changes that coefficient. The resulting coefficient lists are distinct, although the resulting vector remains w.

Conclusion. Every $w \in V$ has infinitely many representations in the dependent spanning family, in particular more than one.

Exercise 4.4.21: Coordinates and basis representations

The basis matrix and its determinant are

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -2 \\ -4 & 9 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = 9 - 8 = 1.$$

Thus the matrix implementing the coordinate map is its inverse:

$$A = P_{\mathcal{B}}^{-1} = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}.$$

Indeed $AP_{\mathcal{B}} = I_2$, so $Ax = AP_{\mathcal{B}}[x]_{\mathcal{B}} = [x]_{\mathcal{B}}$ for every x.

Conclusion. $A = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}$.

Exercise 4.4.22: Coordinates and basis representations

Form $P_{\mathcal{B}} = [b_1 \ \cdots \ b_n]$. Its columns are a basis, so it is invertible. Since $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$, multiply by the inverse to obtain

$$[x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1} x.$$

Conclusion. The implementing matrix is $A = [b_1 \ \cdots \ b_n]^{-1}$, not the basis matrix itself.

Exercise 4.4.23: Coordinates and basis representations

Suppose $[u]_{\mathcal{B}} = [w]_{\mathcal{B}} = (c_1, \ldots, c_n)^T$. By the definition of coordinates, $u = \sum_j c_j b_j$ and $w = \sum_j c_j b_j$. The right sides are identical, so $u = w$.

Conclusion. Equal coordinate outputs imply equal vector inputs; the coordinate map is one-to-one.

Exercise 4.4.24: Coordinates and basis representations

Take any $y = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$ and define $u = y_1 b_1 + \cdots + y_n b_n \in V$. This is a legitimate vector because V is closed under sums and scalar multiplication. The defining representation has weight list y, and its uniqueness gives $[u]_{\mathcal{B}} = y$.

Conclusion. Every vector of $\mathbb{R}^n$ is reached, so the coordinate map is onto.

Exercise 4.4.25: Coordinates and basis representations

Let $C(x) = [x]_{\mathcal{B}}$. It is linear and one-to-one. For any scalars c_j,

$$\sum_j c_j u_j = 0 \quad \Longleftrightarrow \quad C\left(\sum_j c_j u_j\right) = 0 \quad \Longleftrightarrow \quad \sum_j c_j C(u_j) = 0.$$

The forward implication in the first equivalence uses $C(0) = 0$; the reverse uses injectivity. The second equivalence uses linearity. Thus the two homogeneous equations have exactly the same coefficient solutions. One family has only the trivial zero combination if and only if the other does.

Conclusion. The original vectors are independent exactly when their coordinate vectors are independent.

Exercise 4.4.26: Coordinates and basis representations

If $w = \sum_j c_j u_j$, linearity gives $[w]_{\mathcal{B}} = \sum_j c_j [u_j]_{\mathcal{B}}$. Conversely suppose the latter relation holds. Linearity rewrites it as $C(w) = C(\sum_j c_j u_j)$. Injectivity of C then implies $w = \sum_j c_j u_j$. The coefficients themselves are the same in both representations.

Conclusion. Span membership is preserved in both directions by coordinates.

Exercise 4.4.27: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 2 \\ 2 & 0 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a pivot in every column, so the coefficient columns are independent. The coordinate map is injective and linear, so the polynomials are independent as well.

Conclusion. The set is linearly independent.

Exercise 4.4.28: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -2 & 0 & -2 \\ -1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a pivot in every column, so the coefficient columns are independent. The coordinate map is injective and linear, so the polynomials are independent as well.

Conclusion. The set is linearly independent.

Exercise 4.4.29: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(-1)p_1 + (1)p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The set is linearly dependent.

Exercise 4.4.30: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 8 & 9 & 1 \\ -12 & -6 & 6 \\ 6 & 1 & -5 \\ -1 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(1)p_1 + (-1)p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The set is linearly dependent.

Exercise 4.4.31: Coordinates and basis representations

Use the coefficient order $(1, t, t^2)$. Spanning $\mathcal{P}_2$ is equivalent to having a pivot in every row of the coefficient matrix.

(a) Reduce the coefficient matrix.

$$\begin{bmatrix} 1 & -3 & -4 & 1 \\ -3 & 5 & 5 & 0 \\ 5 & -7 & -6 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{5}{4} & -\frac{5}{4} \\ 0 & 1 & \frac{7}{4} & -\frac{3}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Only 2 rows contain pivots, so the set does not span $\mathcal{P}_2$.

(b) Reduce the coefficient matrix.

$$\begin{bmatrix} 0 & 1 & -3 & 2 \\ 5 & -8 & 4 & -3 \\ 1 & -2 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

All three rows contain pivots, so the set spans $\mathcal{P}_2$.

Exercise 4.4.32: Coordinates and basis representations

(a) Check the three coefficient columns.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix}, \qquad \det A = -1 \neq 0.$$

Therefore these columns form a basis of $\mathbb{R}^3$, so the polynomials form a basis of $\mathcal{P}_2$.

(b) Assemble the specified polynomial.

$$q = -p_1 + p_2 + 2p_3 = -(1 + t^2) + (t - 3t^2) + 2(1 + t - 3t^2) = 1 + 3t - 10t^2.$$

Conclusion. $q(t) = 1 + 3t - 10t^2$.

Exercise 4.4.33: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 3 & 5 & 0 & 1 \\ 7 & 1 & 1 & 16 \\ 0 & 0 & -2 & -6 \\ 0 & -2 & 0 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -2 \\ 1 \\ -3 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(-2)p_1 + (1)p_2 + (-3)p_3 + (1)p_4 = 0.$$

Thus the polynomials are dependent.

Conclusion. The polynomials do not form a basis of $\mathcal{P}_3$, since they are dependent.

Exercise 4.4.34: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 5 & 9 & 6 & 0 \\ -3 & 1 & -2 & 0 \\ 4 & 8 & 5 & 0 \\ 2 & -6 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{3}{4} & 0 \\ 0 & 1 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -\frac{3}{4} \\ -\frac{1}{4} \\ 1 \\ 0 \end{bmatrix}$. It gives the polynomial relation

$$(-\frac{3}{4})p_1 + (-\frac{1}{4})p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The polynomials do not form a basis of $\mathcal{P}_3$, since they are dependent.

Exercise 4.4.35: Coordinates and basis representations

Check that the ordered generators are independent.

The coefficient matrix has a pivot in every column, as the reduction below verifies. Hence the generators form a basis of their stated span H.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 11 & 14 & 19 \\ -5 & -8 & -13 \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{11})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ -5 & -8 & -13 \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-5)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (10)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (7)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{11}{18})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (\frac{3}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (\frac{12}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{14}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -\frac{5}{3} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = -\frac{5}{3} \\ c_2 = \frac{8}{3} \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} -\frac{5}{3} \\ \frac{8}{3} \end{bmatrix}$.

Verify membership directly.

The computed weights satisfy $\begin{bmatrix} 11 & 14 \\ -5 & -8 \\ 10 & 13 \\ 7 & 10 \end{bmatrix} \begin{bmatrix} -\frac{5}{3} \\ \frac{8}{3} \end{bmatrix} = \begin{bmatrix} 19 \\ -13 \\ 18 \\ 15 \end{bmatrix}$, so $x \in H$ and the listed coordinates are the unique basis weights.

Exercise 4.4.36: Coordinates and basis representations

Check that the ordered generators are independent.

The coefficient matrix has a pivot in every column, as the reduction below verifies. Hence the generators form a basis of their stated span H.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} -6 & 8 & -9 & 4 \\ 4 & -3 & 5 & 7 \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_1 \leftarrow (-\frac{1}{6})R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 4 & -3 & 5 & 7 \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (4)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-9)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ 0 & -5 & \frac{11}{2} & -14 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (4)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ 0 & -5 & \frac{11}{2} & -14 \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_2 \leftarrow (\frac{3}{7})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & -5 & \frac{11}{2} & -14 \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-5)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & \frac{47}{14} & \frac{47}{7} \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (\frac{7}{3})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & \frac{47}{14} & \frac{47}{7} \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{14}{47})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (-2)R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{3}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{4}{3} & 0 & -\frac{11}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{3}{7})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{4}{3} & 0 & -\frac{11}{3} \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-\frac{4}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = 5 \\ c_3 = 2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$.

Verify membership directly.

The computed weights satisfy $\begin{bmatrix} -6 & 8 & -9 \\ 4 & -3 & 5 \\ -9 & 7 & -8 \\ 4 & -3 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \\ -8 \\ 3 \end{bmatrix}$, so $x \in H$ and the listed coordinates are the unique basis weights.

Exercise 4.4.37: Coordinates and basis representations

What we must show. Convert lattice weights into ordinary spatial coordinates. The basis columns are given in angstroms, so use $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ rather than the inverse direction:

$$x = \begin{bmatrix} \frac{13}{5} & 0 & 0 \\ -\frac{3}{2} & 3 & 0 \\ 0 & 0 & \frac{24}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{4} \\ \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{13}{10} \\ 0 \\ \frac{4}{5} \end{bmatrix}.$$

Coordinatewise, the first value is $2.6(1/2) = 1.3$, the second is the sum of the contributions from the first two basis columns, and the third is 4.8 times the third lattice coordinate.

Conclusion. The standard-coordinate location is $(\frac{13}{10}, 0, \frac{4}{5})^T$ angstroms.

Exercise 4.4.38: Coordinates and basis representations

What we must show. Convert lattice weights into ordinary spatial coordinates. The basis columns are given in angstroms, so use $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ rather than the inverse direction:

$$x = \begin{bmatrix} \frac{13}{5} & 0 & 0 \\ -\frac{3}{2} & 3 & 0 \\ 0 & 0 & \frac{24}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{13}{10} \\ \frac{3}{4} \\ \frac{8}{5} \end{bmatrix}.$$

Coordinatewise, the first value is $2.6(1/2) = 1.3$, the second is the sum of the contributions from the first two basis columns, and the third is 4.8 times the third lattice coordinate.

Conclusion. The standard-coordinate location is $(\frac{13}{10}, \frac{3}{4}, \frac{8}{5})^T$ angstroms.